MILITARY SPECIFICATION

COUPLERS, DIRECTIONAL

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for radio and microwave frequencies directional couplers (see 6.1).

1.2 Military part numbers. The military part number should consist of the letter "M" followed by the basic number of the specification sheet, an assigned dash number (see 3.1), and the letter N or S; where N indicates a nonscreened item and S indicates a screened item. Part numbers without an N or S should be considered nonscreened items.

M15370/14- 001 N or S

- Military designator and specification sheet number
- Dash number designated on specification sheet
- Nonscreened
- Screened

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents should be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ES, 1507 Wilmington Pike, Dayton, OH 45444-5276, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5985

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SPECIFICATIONS

FEDERAL

QQ-M-44	-	Magnesium Alloy Plate and Sheet (AZ31B).
QQ-A-200	-	Aluminum Alloy, Bar, Rod, Shapes, Structural Shapes, Tube and Wire, Extruded.
QQ-A-225	-	Aluminum and Aluminum Alloy Bar, Rod, Wire, or Special Shapes, Rolled, Drawn, or Cold Finished, General Specification for.
QQ-A-250	-	Aluminum and Aluminum Alloy Plate and Sheet, General Specification for.
QQ-S-365	-	Silver Plating, Electrodeposited, General Requirements for.
QQ-C-530	-	Copper-Beryllium Alloy Bar, Rod, and Wire (Copper Alloy Numbers 172 and 173).
QQ-C-533	-	Copper-Beryllium Alloy Strip (Copper Alloy Numbers 170 and 172).
QQ-A-591	-	Aluminum Alloy Die Castings.
QQ-A-596	-	Aluminum Alloy Permanent and Semi-Permanent Mold Castings.
QQ-A-601	-	Aluminum-Alloy Sand Castings.
QQ-B-613	-	Brass, Leaded and Nonleaded, Flat Products (Plate, Bar, Sheet and Strip).
QQ-B-626	-	Brass, Leaded and Nonleaded Rod, Shapes, Forgings and Flat Products TH Finished Edges (Bar and Strip).
QQ-S-763	-	Steel Bar, Wire, Shape and Forging, Corrosion Resisting.
QQ-S-781	-	Strapping, Steel, and Seals.
TT-P-645	-	Primer, Paint, Zinc-Chromate, Alkyd Type.
PPP-B-566	-	Box, Folding, Paperboard.
PPP-B-601	-	Boxes, Wood, Cleated Plywood.
PPP-B-621	-	Box, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Shipping, Fiberboard.
PPP-B-676	-	Boxes Setup.

MILITARY

MIL-P-116	-	Preservation, Methods of.
MIL-P-1144	-	Pipe, Corrosion-Resistant, Stainless Steel, Seamless or Welded.
MIL-M-3171	-	Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on.
MIL-F-3922	-	Flange, Waveguide, General Purpose, General Specification for.
MIL-S-4043	-	Steel, Corrosion-Resisting (Extra Low Carbon Type 304) Plate, Sheet, and Strip.
MIL-B-5087	-	Bonding, Electrical, and Lighting Protection, for Aerospace Systems.
MIL-C-5541	-	Chemical Conversion Coatings on Aluminum Alloys.
MIL-T-10727	-	Tin Plating, Electrodeposited or Hot-Dipped, for Ferrous and Nonferrous Metals.
MIL-E-15090	-	Enamel, Equipment, Light Gray (Formula No. 111).
MIL-P-19468	-	Plastic Rods, Polytetrafluoroethylene, Molded and Extruded.
MIL-P-19834	-	Plate, Identification, Metal Foil, Adhesive Backed.
MIL-C-22751	-	Coating System, Epoxy-Polyamide, Chemical and Solvent Resistant; Process for Application of.
MIL-I-23011	-	Iron Nickel Alloys for Sealing to Glasses and Ceramics.
MIL-C-26074	-	Coating, Electroless Nickel, Requirements for.
MIL-P-27418	-	Plating, Soft Nickel (Electrodeposited, Sulfamate Bath).
MIL-H-28719	-	Header, Hermetically Sealed.
MIL-F-39000	-	Flanges, Waveguide, Ridge, General Specification for.
MIL-C-39012	-	Connectors, Coaxial, Radio Frequency, General Specification for.
MIL-G-45204	-	Gold Plating, Electrodeposited.

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MIL-P-45209 - Palladium Plating, (Electrodeposited).
MIL-I-46058 - Insulating Compound, Electrical (for Coating Printed Circuit Assemblies).
MIL-T-55155 - Terminals, Feedthru (Insulated) and Terminals, Stud (Insulated and Noninsulated).
MIL-C-55302 - Connectors, Printed Circuit Subassembly and Accessories.

(See supplement 1 for list of associated specification sheets.)

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.
FED-STD-595 - Color (Requirements for Individual Color Chip (3X5 Supplements).

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-147 - Palletized Unit Loads.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-794 - Parts and Equipment, Procedures for Packaging and Packing of.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1276 - Leads for Electronic Component Parts.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.
MIL-STD-45662 - Calibration Systems Requirements.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

DEPARTMENT OF THE NAVY

REA 49330 - UG-45/U Connector for Use with 7/8 Coaxial Air Dielectric Line.
REA 49331 - UG-46/U Connector for Use with 7/8 Coaxial Air Dielectric Line.

(Application for copies should be addressed to: Space and Naval Warfare Systems Command, ATTN: SPAWAR 003-1212, Department of the Navy, Washington, DC 20363.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE-STD-287 - Precision Coaxial Connectors.

(Application for copies should be addressed to the Institute of Electrical and Electronics Engineers Headquarters, 345 East 47 Street, New York, NY 10017.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A484	-	Bars, Billets and Forgings, Stainless and Heat-Resisting, General Specification for.
ASTM A582	-	Free-Machining Stainless and Heat-Resisting Steel Bars, Hot-Rolled or Cold-Finished.
ASTM D1457	-	PTFE Molding and Extrusion Materials, Specifications for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual directional coupler requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.1.1 Reference to specification sheet. For the purpose of this specification, when the terms, "as specified", "when specified" or "when applicable" are used without additional reference to a specific location or document, the intended reference shall be to the specification sheet. When the specification sheet does not contain the information, the requirement is not applicable to that specific part number or specification sheet.

3.2 Qualification. Directional couplers covered by specification sheets furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.5 and 6.3).

3.3 Material. The material shall be as specified in table I herein, and in the applicable specification sheets (see 3.1). When a definite material is not specified, a material shall be used which will enable the directional coupler to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Brass. Brass shall conform to QQ-B-626 or QQ-B-613.

3.3.2 Copper alloy. Copper alloy sheet shall conform to QQ-B-613.

3.3.3 Copper, beryllium. Beryllium copper shall conform to QQ-C-530 or QQ-C-533.

3.3.4 Corrosion-resisting steel. Corrosion-resisting steel plates, sheets, and strips shall conform to MIL-S-4043. Corrosion-resisting forgings shall conform to QQ-S-763, ASTM A484 or ASTM A582 and corrosion-resisting steel pipes shall conform to MIL-P-1144.

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3.3.5 Aluminum alloy. Aluminum alloy plates and sheets shall conform to composition 6061 of QQ-A-250, extruded aluminum alloy shall conform to composition 6063 of QQ-A-200 or composition 6061 of QQ-A-250. Aluminum alloy casting shall conform to alloy A360 of QQ-A-591, class 8 of QQ-A-596, alloy 40E of QQ-A-601, or 2011 of QQ-A-225.

3.3.6 Magnesium alloy. Magnesium alloy shall be composition AZ31B, condition H24, in accordance with QQ-M-44. Unless otherwise specified (see 6.2), magnesium couplers shall not be supplied for naval applications.

3.3.7 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MIL-STD-889 shall not be in intimate contact.

3.3.8 Fungus inert material. Material used in the construction of directional couplers shall be fungus inert in accordance with requirement 4 of MIL-STD-454.

3.3.9 Bonding. Bonding shall conform to class R of MIL-B-5087.

3.3.10 Insulating compounds. Insulating compounds shall conform to MIL-I-46058.

3.3.11 Plastics. Plastics shall conform to ASTM D1457 or PTFE of MIL-P-19468.

3.3.12 Rubber. Rubber shall be suitable for use over the specified temperature range.

3.3.13 Iron-nickel alloy. Iron-nickel alloys shall conform to MIL-I-23011.

3.4 Design and construction. Directional couplers shall be of the design, construction, and physical dimensions specified (see 3.1). Directional couplers shall be of the lightest practicable weight consistent with the strength required for sturdiness, safety, and reliability.

3.4.1 Operating frequency range. The frequency range shall be as specified (see 3.1).

3.4.2 Flanges. Flanges shall be designed and manufactured as to provide the mating characteristics of the flange specified in accordance with MIL-F-3922 or MIL-F-39000.

3.4.3 RF connectors. The connectors shall be as specified (see 3.1). The material and gauging for receptacle connectors shall conform to the requirements of MIL-C-39012 or Drawing REA 49330 or Drawing REA 49331, whichever is applicable. When specified, precision connectors shall be in accordance with IEEE-STD-287.

3.4.3.1 Connector metal parts. Unless otherwise specified (see 3.1), the male center contact pins shall be captivated and made of corrosion-resisting steel or beryllium copper. Corrosion-resisting steel pins shall be type 302 or type 304 in accordance with QQ-S-763, or type 303 in accordance with ASTM A582. Beryllium copper pins shall conform to QQ-C-530 and shall be silver plated in accordance with QQ-S-365 or gold plated in accordance with MIL-G-45204, type II, class 1. The female center contact pins shall be captivated and made of beryllium copper in accordance with QQ-C-530 and silver plated in accordance with QQ-S-365 or gold plated in accordance with MIL-G-45204, type II, class 1.

3.4.4 Printed-circuit connectors. Printed-circuit connectors shall conform to MIL-C-55302.

3.4.5 External leads. External lead connections shall be a chemical composition conforming to MIL-STD-1276 or MIL-I-23011 and shall be solderable, unless otherwise specified (see 3.1).

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3.4.6 Socket pins. Socket pins shall conform to MIL-H-28719.

3.4.7 Terminals. Terminals shall conform to MIL-T-55155.

3.4.8 Headers. Headers shall conform to MIL-H-28719.

3.4.9 Epoxy. The use of epoxy shall be restricted to the following:

a. Impregnation of couplers to seal porous casting and braze joints after brazing.

b. Repair of pin-hole type leaks.

c. Component staking.

d. Cavity encapsulation.

e. Printed-circuit (PC) board and component attachment.

f. PC board material.

g. Capture of connector parts.

h. Sealing for moisture and EMI seal.

3.4.10 Housing. The housing shall be sealed to prevent entry of moisture and leakage of radiated electromagnetic interference (EMI).

3.4.11 Finish. Unless otherwise specified (see 3.1), the finish shall be as specified in 3.4.11.1 through 3.4.11.3.

3.4.11.1 RF mating surfaces. Mating surfaces shall be finished in gold, silver, or tin in accordance with MIL-G-45204, QQ-S-365, MIL-T-10727 or passivated stainless steel. The minimum thickness for gold and nickel plating shall be 10 microinches and 30 microinches, respectively.

3.4.11.2 Plating. All metal parts of directional couplers which are not corrosion-resistant shall have a plating of silver, type I or II, grade A, palladium, gold, nickel, or tin, in accordance with QQ-S-365, MIL-P-45209, MIL-G-45204, MIL-C-26074, or MIL-T-10727 whichever is applicable (see 3.1). The minimum thickness for gold and nickel plating shall be 10 microinches and 30 microinches, respectively. Aluminum alloy surfaces shall be chemically treated in accordance with MIL-C-5541. Magnesium-alloy surfaces shall be given a chrome-nickel treatment in accordance with type I of MIL-M-3171.

3.4.11.3 External finish. External finish shall be applied to the directional couplers regardless of plating or chemical treatment, except that the mating surfaces shall not be coated. The primer coat shall be zinc chromate in accordance with TT-P-645. Two finish coats of enamel, in accordance with type III, class 2, of MIL-E-15090, shall be applied. Optionally, epoxy paint system MIL-C-22751 may be used. External coating shall be applied as continuous film.

3.4.12 Weight. The weight of the directional coupler shall be as specified (see 3.1).

3.4.13 Temperature range. The operating and nonoperating temperature ranges shall be as specified (see 3.1).

3.4.14 Pre-seal bake. Hermetically sealed directional couplers shall have a pre-seal bake in an inert atmosphere at the maximum specified storage or operating temperature, whichever is greater, for a minimum of 16 hours. There shall be a direct transfer to the seal chamber with an inert atmosphere that has a monitored moisture content of less than 0.1 percent.

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3.4.15 Connection cap. All connections that are not normally sealed shall be capped with push-on plastic caps to prevent both damage and the entrance of moisture and foreign material during shipment and storage.

3.4.16 Threaded parts. All screw threads used in the construction of directional couplers shall be in accordance with FED-STD-H28.

3.4.17 RF input power to primary. Directional couplers shall be designed to meet the electrical requirements with the specified RF input power to the primary.

3.5 Performance.

3.5.1 Screening. All screened directional couplers produced to this specification shall be screened in accordance with table IV. After screening, directional couplers shall be subjected to group A inspection (see table V).

3.5.2 Coupling. When directional couplers are tested as specified in 4.7.3, the measured value of coupling shall be within the tolerance specified for the nominal value. This specified tolerance includes the allowed variation of coupling over the specified frequency ranges.

3.5.3 Coupling variation (frequency sensitivity). When directional couplers are tested as specified in 4.7.4, the plus and minus variation in coupling (in decibels (dB)) over the specified frequency range shall not exceed the value specified. Plus and minus values are determined from the measured coupling level.

3.5.4 Effective directivity. When directional couplers are tested as specified in 4.7.5, the effective directivity (in dB) over the specified frequency range shall not exceed the value specified.

3.5.5 Insertion loss (see 6.4.2). When directional couplers are tested as specified in 4.7.6, the insertion loss (excluding coupling power loss) shall not exceed the value specified.

3.5.6 Voltage standing wave ratio (VSWR).

3.5.6.1 Primary line (see 6.4.3). When directional couplers are tested as specified in 4.7.7.1, the VSWR over the specified frequency range shall not exceed the value specified.

3.5.6.2 Secondary line (see 6.4.4). When directional couplers are tested as specified in 4.7.7.2, the VSWR over the specified frequency range shall not exceed the value specified.

3.5.7 Power dissipation of secondary-line termination. When directional couplers are tested as specified in 4.7.8, the VSWR of the secondary line at mid-frequency (see 6.4.5) of the specified frequency range shall be as specified in 3.5.6.2.

3.5.8 Coaxial connector wear resistance. When directional couplers are tested as specified in 4.7.9, there shall be no damage to the connectors that will cause an electrical failure. During and after cycling, neither lubrication or removal of excess material shall be permitted. After this test, directional couplers shall meet the requirements of 3.5.5 and 3.5.6.

3.6 Solderability. When directional couplers with solderable connections are tested as specified in 4.7.10, there shall be no evidence of pin holes and blistering.

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3.7 Resistance to soldering heat. When directional couplers with solderable connections are tested as specified in 4.7.11, there shall be no damage to the directional coupler or to the terminal insulator that will cause electrical failure. Chipping of the terminal insulator shall not be cause for failure unless the chipping extends to the outer periphery. After this test, directional couplers shall meet the requirements of 3.5.6.

3.8 Terminal strength-lead integrity. When directional couplers with terminals or leads are tested as specified in 4.7.12, there shall be no evidence of a broken terminal or lead elongation greater than 1/2 of the thread pitch, or breakage, loosening or relative motion between the terminal and the directional coupler body when viewed through a magnification of at least 10X. Any of these shall be considered a failure.

3.9 Resistance to solvents. When directional couplers are tested as specified in 4.7.13, there shall be no evidence of illegible marking, mechanical damage, or deterioration of material or finishes to the extent that they can be readily identified from a distance of at least 6 inches with normal room lighting and without the aid of magnification or with a viewer having a magnification no greater than 3X.

3.10 Thermal shock. When directional couplers are tested as specified in 4.7.14, there shall be no evidence of physical damage. Upon completion of this test, directional couplers shall meet the requirements of 3.5.2, 3.5.4, 3.5.5, and 3.5.6.

3.11 Vibration. When directional couplers are tested as specified in 4.7.15, there shall be no evidence of physical damage.

3.12 Shock. When directional couplers are tested as specified in 4.7.16, there shall be no evidence of physical damage. Upon completion of this test, directional couplers shall meet the requirements of 3.14 or 3.15, as applicable.

3.13 Acceleration. When directional couplers are tested as specified in 4.7.17, there shall be no evidence of physical damage. Upon completion of this test, directional couplers shall meet the requirements of 3.5.2, 3.5.4, 3.5.5, and 3.5.6.

3.14 Seal. When hermetically sealed directional couplers are tested as specified in 4.7.18, the applicable amount of leakage rate shall not be exceeded nor shall there be evidence of physical damage. Upon completion of this test, directional couplers shall meet the requirements of 3.5.2, 3.5.4, 3.5.5, and 3.5.6.

3.15 Pressurization. When waveguide type directional couplers are tested as specified in 4.7.19, there shall be no leakage, as detected by the continuous formation of escaping air bubbles. Upon completion of this test, directional couplers shall meet the requirements of 3.5.2, 3.5.4, 3.5.5, and 3.5.6.

3.16 Barometric pressure. When directional couplers are tested as specified in 4.7.20, there shall be no evidence of physical damage. During this test, directional couplers shall meet the requirements of 3.5.2, 3.5.4, 3.5.5, and 3.5.6.

3.17 Moisture resistance. When directional couplers are tested as specified in 4.7.21, there shall be no destructive corrosion. Destructive corrosion shall be construed as any type of corrosion which in any way interferes with mechanical performance or appearance.

3.18 Salt spray. When directional couplers are tested as specified in 4.7.22, there shall be no evidence of warping, cracking, peeling or corrosion that has passed through the plating and exposed the base metal, or any lead breakage when viewed through a magnification of at least 10X. Any of these shall be considered a directional coupler failure.

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3.19 Electromagnetic interference (EMI). When coaxial or waveguide directional couplers are tested as specified in 4.7.23, the RF leakage from the directional coupler shall be at least 65 dB below the incoming signal level.

3.20 Life. When directional couplers are tested as specified in 4.7.24, there shall be no evidence of damage. Upon completion of this test, directional couplers shall meet the requirements of 3.5.2 through 3.5.7.

3.21 Marking. Directional couplers shall be marked in accordance with MIL-STD-1285 with the military part number, manufacturer's source code or logo, the coupling value (in dB), date code, and serialization in the location specified. On unidirectional couplers, an arrow shall be placed so as to point in the direction in which incident power flows. On bidirectional couplers which have two different couplings, the coupling of each secondary line shall be indicated separately in a manner that will make clear which coupling is associated with each secondary line. Bidirectional couplers shall be marked to indicate which of the secondary lines has a nominal response to each direction of incident power flows. Marking shall be accomplished by reverse etching on metal identification plates, by engraving, by photo etching in accordance with MIL-P-19834, with permanent ink stamping in accordance with MIL-STD-130, or silk screen. The marking shall be placed on the directional coupler.

3.21.1 Date code. Directional couplers shall be marked by a unique code to identify the period during which they were manufactured. The first two numbers in the code shall be two digits of the number of the year, and the third and fourth numbers shall be two digits indicating the calendar week of the year. When the number of the week is a single digit, it shall be preceded by a zero reading from left to right or from top to bottom, the code number shall designate the year and week, in that order. The date code shall not be altered or removed from the directional coupler. A new date code followed by the letter R shall be marked on all reworked directional couplers.

3.21.2 Serialization. Each directional coupler shall be marked with a unique serial number assigned consecutively within the inspection lot allowing traceability of the directional coupler.

3.22 Workmanship. Directional couplers shall be manufactured and processed in a careful and workmanlike manner in accordance with MIL-STD-454, requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Materials inspection (see 4.3).

b. Qualification inspection (see 4.5).

c. Quality conformance inspection (see 4.6).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table I, used in fabricating the directional coupler, are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

TABLE I. Materials inspection.

Material	Requirement paragraph	Applicable specification
Brass	3.3.1	QQ-B-613, QQ-B-626
Copper alloy	3.3.2	QQ-B-613
Copper, beryllium	3.3.3	QQ-C-530, QQ-C-533
Corrosion-resisting steel	3.3.4	MIL-P-1144, QQ-S-763, ASTM A484, ASTM A582, MIL-S-4043
Aluminum alloy	3.3.5	QQ-A-200, QQ-A-250, QQ-A-225, QQ-A-596, QQ-A-601, QQ-A-591
Magnesium alloy	3.3.6	QQ-M-44
Dissimilar metals	3.3.7	MIL-STD-889
Fungus inert material	3.3.8	MIL-STD-454
Bonding	3.3.9	MIL-B-5087
Insulating compounds	3.3.10	MIL-I-46058
Plastics	3.3.11	ASTM D1457, MIL-P-19468
Iron-nickel alloy	3.3.13	MIL-I-23011
Flanges	3.4.2	MIL-F-3922, MIL-F-39000
Connectors	3.4.3, 3.4.4	MIL-C-39012, REA 49330, REA 49331, QQ-S-763, ASTM A582, IEEE-STD-287, MIL-C-55302
External leads	3.4.5	MIL-STD-1276, MIL-I-23011
Socket pins	3.4.6	MIL-H-28719
Terminals	3.4.7	MIL-T-55155
Headers	3.4.8	MIL-H-28719
Finish	3.4.11	MIL-G-45204, MIL-P-27418, QQ-S-365, MIL-T-10727, MIL-P-45209, MIL-T-26074, MIL-C-5541, TT-P-645, MIL-E-15090, MIL-M-3171, MIL-C-22751
Threaded parts	3.4.16	FED-STD-H28

4.4.1 Test method variation. Variation from the specified test methods used to verify the electrical parameters are allowed provided that it is demonstrated to the preparing activity or their agent that such variations in no way relax the requirements of this specification and that they are approved before testing is performed. For proposed test variations a test method comparative error analysis shall be made available for checking by the preparing activity or their agent.

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4.5 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3.1) on sample units produced with equipment and procedures normally used in production. Qualification obtained for each group (see appendix) shall constitute qualification for all couplers with characteristics for that group.

4.5.1 Sample size. Four samples of the same part number shall be subjected to qualification inspection.

4.5.2 Inspection routine. The samples shall be subjected to the qualification inspections specified in table II, in the order shown. All units shall be subjected to the inspections of group I. The samples shall then be divided into two groups of two units each (see 4.5.1). The sample units shall then be subjected only to the inspections indicated for their particular group.

4.5.3 Failures. One or more failures shall be cause for refusal to grant qualification approval. A failure shall be anything that does not meet the requirements of the specification.

4.5.4 Disposition of qualification sample units. Sample units which have been subjected to qualification testing shall not be delivered on any contract or purchase order. The Government reserves the right to retain the sample units or to require the contractor to furnish the sample units with the qualification inspection report.

4.5.5 Retention of qualification. To retain qualification, the contractor shall forward a report at 24-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

a. A summary of the results of the tests performed for inspection of product for delivery (group A), indicating as a minimum, the number of couplers that have passed and the number that have failed. The results of tests of all reworked couplers shall be identified and accounted for.

b. A summary of the results of tests performed for periodic inspection (see table III), including the number and mode of failures. The summary shall include results of all periodic inspection tests performed. If the summary of test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 60 days after the end of each 24-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 24-month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the couplers. If during the two consecutive reporting periods there has been no production, the manufacturers may be required, at the discretion of the qualifying activity, to submit their qualified products to testing in accordance with the qualification inspection requirements and the reason for no production.

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TABLE II. Qualification inspection.

Inspection	Requirement paragraph	Test method paragraph
Group I - all samples		
Screened per table IV (for screened directional coupler only)	3.5.1	4.7.2
Visual and mechanical inspection 1/	3.1, 3.3, 3.4, 3.21 and 3.22	4.7.1
Coupling	3.5.2	4.7.3
Coupling variation	3.5.3	4.7.4
Effective directivity	3.5.4	4.7.5
Insertion loss	3.5.5	4.7.6
VSWR	3.5.6	4.7.7
Power dissipation of secondary-line termination	3.5.7	4.7.8
Coaxial connector wear resistance 2/	3.5.8	4.7.9
Group II - 2 sample units		
Solderability 3/	3.6	4.7.10
Resistance to soldering heat 3/	3.7	4.7.11
Terminal strength-lead integrity 4/	3.8	4.7.12
Resistance to solvents	3.9	4.7.13
Thermal shock	3.10	4.7.14
Coupling	3.5.2	4.7.3
Effective directivity	3.5.4	4.7.5
Insertion loss	3.5.5	4.7.6
VSWR	3.5.6	4.7.7
Vibration (high frequency)	3.11	4.7.15.1
Shock	3.12	4.7.16
Seal 5/	3.14	4.7.18
Pressurization 6/	3.15	4.7.19
Acceleration	3.13	4.7.17
Coupling	3.5.2	4.7.3
Effective directivity	3.5.4	4.7.5
Insertion loss	3.5.5	4.7.6
VSWR	3.5.6	4.7.7
Electromagnetic interference	3.19	4.7.23
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.21 and 3.22	4.7.1
Group III - 2 sample units		
Vibration (random)	3.11	4.7.15.2
Barometric pressure	3.16	4.7.20
Coupling	3.5.2	4.7.3
Effective directivity	3.5.4	4.7.5
Insertion loss	3.5.5	4.7.6
VSWR	3.5.6	4.7.7
Moisture resistance	3.17	4.7.21
Salt spray	3.18	4.7.22
Life	3.20	4.7.24
Coupling	3.5.2	4.7.3
Coupling variation	3.5.3	4.7.4
Effective directivity	3.5.4	4.7.5
Insertion loss	3.5.5	4.7.6
VSWR	3.5.6	4.7.7
Power dissipation of secondary-line termination	3.5.7	4.7.8
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.21 and 3.22	4.7.1

See footnotes on next page.

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1/ For nonscreened directional couplers only.
2/ Coaxial types of directional couplers.
3/ For directional couplers with solderable leads or terminals.
4/ Directional couplers with leads or terminals.
5/ For all types of directional couplers, with the exception of the waveguide type.
6/ For waveguide type of directional couplers.

4.5.6 Periodic inspection. Periodic inspection shall consist of the inspections specified in table III, in the order shown. The sample shall be divided into two groups of two units each. The units shall then be subjected only to the inspections indicated for their particular group. Periodic inspection shall be made on sample units selected from inspection lots which have passed the group A inspection.

Except where the results of these inspections show noncompliance with the applicable requirements (see 4.5.6.4), delivery of products which have passed the screening and group A inspection (screened items only) or group A inspection (nonscreened items only) shall not be delayed pending the results of these periodic inspections.

4.5.6.1 Sampling plan. Four sample units (see 4.5.2) shall be selected every 24 months. The first inspection shall be 24 months after the date of notification of qualification.

4.5.6.2 Failures. If one or more sample units fail to pass periodic inspection, the sample shall be considered to have failed.

4.5.6.3 Disposition of sample units. Sample units which have been subjected to periodic inspection shall not be delivered on the contract.

4.5.6.4 Noncompliance. If a sample fails to pass periodic inspection, the manufacturer shall notify the qualifying activity and cognizant inspection activity of such failure and take corrective action on the material or processes, or both, as warranted, and on all units of products which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, periodic inspection shall be repeated on additional sample units (all inspections, or the inspection which the original sample failed) at the option of the qualifying activity. Group A inspection may be reinstituted; however, final acceptance shall be withheld until the periodic inspection has shown that corrective action was successful. In the event of failure after inspection, information concerning the failure and the corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

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TABLE III. Periodic inspection.

Inspection	Requirement paragraph	Test method paragraph
Group I - 2 sample units		
Solderability 1/	3.6	4.7.10
Resistance to soldering heat 1/	3.7	4.7.11
Terminal strength-lead integrity 2/	3.8	4.7.12
Resistance to solvents	3.9	4.7.13
Thermal shock	3.10	4.7.14
Coupling	3.5.2	4.7.3
Effective directivity	3.5.4	4.7.5
Insertion loss	3.5.5	4.7.6
VSWR	3.5.6	4.7.7
Vibration (high frequency)	3.11	4.7.15.1
Shock	3.12	4.7.16
Seal 3/	3.14	4.7.18
Pressurization 4/	3.15	4.7.19
Acceleration	3.13	4.7.17
Coupling	3.5.2	4.7.3
Effective directivity	3.5.4	4.7.5
Insertion loss	3.5.5	4.7.6
VSWR	3.5.6	4.7.7
Electromagnetic interference	3.19	4.7.23
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.21 and 3.22	4.7.1
Group II - 2 sample units		
Vibration (random)	3.11	4.7.15.2
Barometric pressure	3.16	4.7.20
Coupling	3.5.2	4.7.3
Effective directivity	3.5.4	4.7.5
Insertion loss	3.5.5	4.7.6
VSWR	3.5.6	4.7.7
Moisture resistance	3.17	4.7.21
Salt spray	3.18	4.7.22
Life	3.20	4.7.24
Coupling	3.5.2	4.7.3
Coupling variation	3.5.3	4.7.4
Effective directivity	3.5.4	4.7.5
Insertion loss	3.5.5	4.7.6
VSWR	3.5.6	4.7.7
Power dissipation of secondary-line termination	3.5.7	4.7.8
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.21 and 3.22	4.7.1

1/ For directional couplers with solderable leads or terminals.

2/ Directional couplers with leads or terminals.

3/ For all types of directional couplers, with the exception of the waveguide type.

4/ For waveguide type of directional couplers.

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4.6 Quality conformance inspection.

4.6.1 General. Quality conformance inspection shall consist of group A inspection and screening (when applicable).

4.6.1.1 Inspection lot. An inspection lot shall consist of all directional couplers with the same part number produced under essentially the same conditions, and offered for inspection at one time.

4.6.1.2 Screening. Screening shall consist of the inspection and tests specified in table IV, in the order shown. Directional couplers shall pass screening before being subjected to group A inspection.

TABLE IV. Screening.

Inspection	Requirement paragraph	Test method paragraph
Visual and mechanical inspection	3.5.1	4.7.2.1
Thermal shock	3.5.1	4.7.2.2
Radiographic inspection	3.5.1	4.7.2.3

4.6.1.3 Group A inspection. Group A inspection shall consist of the inspection and tests specified in table V, in the order shown.

4.6.1.4 One-hundred-percent inspection. All screened directional couplers shall be subjected to screening and group A inspection. All nonscreened directional couplers shall be subjected to group A inspection. Defective units shall be individually rejected.

4.6.1.5 Rejected items. If an inspection item is rejected, the contractor may rework it to correct the defect and resubmit for reinspection. Such items shall be separate from new items and shall be clearly identified as reinspected items.

4.6.1.6 Test data. Data shall be taken and recorded for all tests performed on an individual item and sent to the acquiring activity. The manner of performing measurements and data sheets shall be included and shipped in the same container as the directional coupler. No classified information shall appear on the data sheet.

TABLE V. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph
Visual and mechanical inspection	3.1, 3.3, 3.4, 3.21 and 3.22	4.7.1
Coupling	3.5.2	4.7.3
Coupling variation	3.5.3	4.7.4
Effective directivity	3.5.4	4.7.5
Insertion loss	3.5.5	4.7.6
VSWR	3.5.6	4.7.7
Power dissipation of secondary-line termination	3.5.7	4.7.8

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4.6.2 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with the groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.7 Methods of inspection and test.

4.7.1 Visual and mechanical inspection (see 3.1, 3.4, 3.21 and 3.22). Directional couplers shall be inspected to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements.

4.7.2 Screening (see 3.5.1). Directional couplers shall be screened as specified in 4.7.2.1 through 4.7.2.3.

4.7.2.1 Visual and mechanical inspection. Visual and mechanical inspection of directional couplers shall be as specified in 4.7.1.

4.7.2.2 Thermal shock. With the connections uncovered, directional couplers shall be tested in accordance with method 107 of MIL-STD-202. The following details and exception shall apply:

a. Mounting: When applicable, directional couplers may be mounted on a heat sink.

b. Test condition: B, except the temperature extremes shall be those specified (see 3.1), and the number of cycles shall be 10. Place directional couplers in the cold chamber for one-half hour. Directional couplers shall be so positioned that they are exposed to freely circulating chamber air. Remove directional couplers from the cold chamber and place in the hot chamber for one-half hour. Transfer shall take place within five minutes of removal from the cold chamber. One cycle consists of room temperature to cold, to room temperature, to hot, and back to room temperature.

4.7.2.3 Radiographic inspection. Directional couplers shall be inspected in accordance with method 209 of MIL-STD-202, the following details shall apply:

a. A three-dimensional image quality indicator, constructed of, and containing materials which are compatible with the directional coupler being x-rayed, shall be available for simultaneous exposure with the directional coupler. The image indicator shall utilize slots, wires, and particles of sizes down to and including major dimensions equal to or less than 0.001 inch for monolithic microelectronic type devices, 0.005 inch for non-microelectronic type devices, provided spacing between nonconnected conductors is equal to or greater than 0.010 inch.

b. Two views shall be required, at 90° to each other, and both perpendicular to the major (long) axis of the directional coupler.

c. The radiographs shall be inspected under variable lighting conditions and magnification of 1X to 7X for the defects specified in the "EVALUATIONS" paragraph of MIL-STD-202, method 209.

d. The directional couplers that have been x-rayed and found acceptable shall be identified with a blue dot. The blue dot shall be approximately 1/16 inch in diameter. The color selected from FED-STD-595 shall be any shade between 15102-15123 and 25102-25109. The dot shall be placed so that it is readily visible but shall not obliterate other directional coupler markings.

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e. Serialization of directional couplers is required and must be correlatable to the film views. One film copy is required to be shipped with the directional couplers. The manufacturer is not required to retain film copies. One copy of a report is required for each lot shipped indicating the number of directional couplers radiographically inspected, the number found acceptable, and the number rejected.

4.7.3 Coupling (see 3.5.2). The coupling of directional couplers shall be determined by obtaining the midpoint between the maximum and minimum coupling over the specified frequency range. The coupling of unidirectional couplers shall be determined as the ratio, expressed in dB, of the power input to the primary line to the power available at the output of the secondary line, with the output end of the primary line properly terminated. The coupling of bidirectional couplers shall be determined separately for each secondary line.

4.7.4 Coupling variation (see 3.5.3). The coupling variation of directional couplers shall be determined over the specified frequency range (see 3.1). The coupling variation of unidirectional couplers shall be determined by taking the difference between the maximum and the minimum coupling over the specified frequency range. Such difference shall be taken as a positive number or zero. The coupling variation of bidirectional couplers shall be determined separately for each secondary line.

4.7.5 Effective directivity (see 3.5.4). The effective directivity of directional couplers shall be determined over the specified frequency (see 3.1). The effective directivity of unidirectional couplers shall be computed from the ratio, taken as greater than unity and expressed in dB, of the available power at the output of the secondary line for the two directions of excitation, at equal power levels, of the primary line; the secondary line shall be terminated in a matched detector. The effective directivity of bidirectional couplers shall be determined separately for each secondary line. At the option of the Government (see 6.2), the alternate test specified in 4.7.5.1 may be used.

4.7.5.1 Alternate test. The effective directivity of directional couplers shall be determined over the specified frequency range. The effective directivity shall be computed from the difference between measurements of minimum and maximum outputs of a secondary line, when a short circuit is displaced through one-half a wave-length in the primary line. The generator at the primary line shall be matched to the primary line. The effective directivity of bidirectional couplers shall be determined separately for each secondary line with the remaining secondary lines terminated by an impedance equal to the characteristic impedance of the secondary line.

4.7.6 Insertion loss (see 3.5.5). The insertion loss shall be measured by a substitution technique using radio, audio, or intermediate frequency (RF, AF, or IF). An average of three test runs shall be used as a final result.

4.7.7 VSWR (see 3.5.6).

4.7.7.1 Primary line (see 3.5.6.1). The VSWR of directional couplers shall be measured over the specified frequency range (see 3.1). The VSWR of unidirectional couplers shall be measured at the input end of the primary line, with the output end of the primary line and the secondary line terminated in matched loads. The VSWR of bidirectional couplers shall be measured at one end of the primary line, with the other end terminated in a matched load; the secondary lines shall be terminated in matched loads.

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4.7.7.2 Secondary line (see 3.5.6.2). The VSWR of directional couplers shall be measured over the specified frequency range (see 3.1). The VSWR of unidirectional couplers shall be measured at the output of the secondary line with both ends of the primary line terminated in matched loads. The VSWR of each secondary line of bidirectional couplers shall be measured at the output of the secondary line with the other secondary line and both ends of the primary line terminated in matched loads.

4.7.8 Power dissipation of secondary-line termination (see 3.5.7). The power dissipation of the secondary-line termination shall be determined over the specified frequency range (see 3.1). The power (P_i') to be applied shall be determined from the following formula:

$$P_i' = \frac{P_o}{\text{antilog}_{10}\left(\frac{\alpha}{10}\right)}$$

Where:

P_i' = Power.
P_o = Continuous-wave power rating of primary line (see 6.2).
α = Coupling in dB.

The power (P_i') shall be applied for a period of one hour to the output of the secondary line, with both ends of the primary line terminated in matched loads. Within 3 minutes after the 1-hour period, the VSWR of the secondary line shall be measured as specified 4.7.7.2, at mid-frequency in the specified frequency range. At the option of the Government, the alternate test specified in 4.7.8.1 may be used.

4.7.8.1 Alternate test. The power dissipation of the secondary-line termination shall be determined over the specified frequency range. Rated continuous-wave power (see 6.2) shall be applied for a period of one hour to the input end of the primary line, with the output end of the primary line and the output of the secondary line terminated in matched loads. After the 1-hour period, the VSWR of the secondary line shall be measured as specified in 4.7.7.2, at mid-frequency in the specified frequency range (see 3.1).

4.7.9 Coaxial connector wear resistance (see 3.5.8). The connectors shall be subjected to 500 cycles of connection and disconnection. A cycle shall consist of a firm connection made to the connectors of the directional coupler with the coupling means tightened to normal tightness and the connectors then completely disconnected and removed from the test circuit.

4.7.10 Solderability (see 3.6). The terminals of the directional coupler shall be tested in accordance with method 208 of MIL-STD-202.

4.7.11 Resistance to soldering heat (see 3.7). Directional couplers shall be tested in accordance with method 210 of MIL-STD-202. The following details and exceptions shall apply:

a. Special preparation: The terminals shall not have been soldered previously.

b. Depth of immersion in the molten solder: To a point 1/16 inch + 1/32 inch, -0 inch from the body.

c. Test condition: A.

d. Cooling time: Stabilize to +25°C.

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4.7.12 Terminal strength-lead integrity (see 3.8). Directional couplers shall be tested as specified in 4.7.12.1 or 4.7.12.2.

4.7.12.1 Terminal strength. Directional couplers with terminals shall be tested in accordance with method 211 of MIL-STD-202, test condition A, applied force 1.5 pounds.

4.7.12.2 Lead integrity. Directional couplers with leads shall be tested in accordance with method 211 of MIL-STD-202, test condition C. The applied force shall be 8 ounces ±0.5 ounce. For leads with a section modulus equal to or less than that of a lead with a cross-section of 0.006 X 0.20, the force shall be 3 ounces ±0.3 ounce.

4.7.13 Resistance to solvents (see 3.9). Directional couplers shall be tested in accordance with method 215 of MIL-STD-202. All portions of the directional coupler shall be brushed.

4.7.14 Thermal shock (see 3.10). Directional couplers shall be tested as specified in 4.7.2.2. After this test, the requirements of 3.5.2, 3.5.4, 3.5.5, and 3.5.6 shall be measured at the inspection conditions specified in 4.4.

4.7.15 Vibration (see 3.11). Directional couplers shall be tested as specified in 4.7.15.1 or 4.7.15.2. When no method is specified, directional couplers shall be tested as specified in 4.7.15.1.

4.7.15.1 High frequency (nonoperating) (see 3.11). Directional couplers shall be tested in accordance with method 204 of MIL-STD-202. The following details shall apply:

a. Mounting of specimens: Couplers shall be attached to the vibration table by means of clamps. A clamp shall be placed around the center of each coupler, and no part of the coupler shall touch any object other than the clamp.

b. Test condition: D.

c. Resonance: There shall be no resonances at or below 40 hertz (Hz).

4.7.15.2 Random vibration. Directional couplers shall be tested in accordance with method 214 of MIL-STD-202, test condition F for 15 minutes duration, mounted by normal means.

4.7.16 Shock (see 3.12). Directional couplers shall be tested as specified in 4.7.16.1, 4.7.16.2, 4.7.16.3, or 4.7.16.4.

4.7.16.1 Coaxial and dual-in-line types. Directional couplers shall be tested in accordance with method 213 of MIL-STD-202, test condition H, mounted by normal means.

4.7.16.2 Header and TO types. Directional couplers shall be tested in accordance with method 213 of MIL-STD-202, test condition C, mounted by normal means.

4.7.16.3 Flat pack type. Directional couplers shall be tested in accordance with method 213 of MIL-STD-202, test condition F, mounted by normal means.

4.7.16.4 Waveguide type. Directional couplers shall be tested in accordance with method 213 of MIL-STD-202, test condition I, mounted by normal means.

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4.7.17 Acceleration (see 3.13). Unless otherwise specified (see 3.1), directional couplers shall be tested in accordance with method 212 of MIL-STD-202. The following details shall apply:

a. Mounting: Mounted by normal means.

b. Test condition: A, with 100 grams (g) level.

4.7.18 Seal.

4.7.18.1 Hermetic seal (see 3.5.1.1). Hermetically sealed items shall be tested in accordance with method 112 of MIL-STD-202, test condition D.

4.7.18.2 O-ring seal, solder seal, or encapsulated seal (see 3.5.1.2). O-ring sealed, solder sealed, or encapsulated sealed items shall be tested in accordance with method 104 of MIL-STD-202, test condition B.

4.7.18.3 Cover seal (see 3.5.1.3). Cover sealed items shall be tested in accordance with method 103 of MIL-STD-202. The following shall apply:

a. Magnitude of test voltage: As specified (see 3.1).

b. Nature of potential: As specified (see 3.1).

c. Points of application: As specified (see 3.1).

d. Minimum kilovoltampere rating of high voltage source: As specified (see 3.1).

e. Limiting value of surge current: As specified (see 3.1).

f. Maximum leakage current: As specified (see 3.1).

g. Test condition letter: As specified (see 3.1).

4.7.19 Pressurization (see 3.15). Waveguide-type directional couplers shall be subjected to an internal gas pressure of 30 pounds per square inch for at least 20 seconds while immersed in water at approximately 20°C. The ends of the primary line shall be appropriately sealed.

4.7.20 Barometric pressure (see 3.16). Directional couplers shall be tested in accordance with method 105 of MIL-STD-202, test condition D.

4.7.21 Moisture resistance (see 3.17). Directional couplers shall be tested in accordance with method 106 of MIL-STD-202, step 7B shall be performed.

4.7.22 Salt spray (see 3.18). Directional couplers shall be tested in accordance with method 101 of MIL-STD-202, test condition B. The RF ports of the directional coupler shall be sealed for this test. After this test, the directional couplers are allowed to be washed and dried before being inspected.

4.7.23 Electromagnetic interference (see 3.19). The swept frequency measurement shall be made covering the total frequency range in steps not exceeding an octave band (the appropriate stub antenna should be tuned to a quarter wave at midoctave).

a. Place a fixed amount of RF power in a transmission line in series with available attenuator and spectrum analyzer.

b. Place specified value of attenuation (see 3.19 or 3.1) in the line and note the difference in reading on the spectrum analyzer.

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c. Reset the attenuator to zero and place a λ/4 stub (at midband) at one end of the flexible coaxial cable and connect to the spectrum analyzer.

d. Place the test attenuator in a transmission line properly terminated and "SNIFF" the coupler with the λ/4 stub. The λ/4 stub should come as close to the coupler as possible without touching. Particular attention shall be given to RF connections.

e. Any attenuation in excess of the value specified in 4.7.23b shall be cause for rejecting the coupler.

4.7.24 Life (see 3.20). Directional couplers shall be placed within a temperature chamber. The chamber temperature shall be raised so that the measurement temperature is equal to the highest specified operating temperature (see 3.1). Directional couplers shall remain at this temperature for a period of 1,000 hours. At the end of this time, directional couplers shall be removed from the temperature chamber and allowed to cool down. Directional couplers shall be considered cooled when their body temperature is equal to the room temperature.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Directional couplers shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Directional couplers shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packs. Each directional coupler shall be individually unit packed in accordance with submethod IA-15 of MIL-P-116 insuring compliance with the applicable requirements of that specification. The container within the unit pack shall conform to variety 2 of PPP-B-566, PPP-B-676, or PPP-B-636, class weather resistant.

5.1.1.5 Intermediate packs. Intermediate packs are not required.

5.1.2 Level B. The requirements for level B shall be as specified for level A except that any variety or class of the containers specified may be used (see 5.1.1.4).

5.1.3 Level C. The level C preservation of directional couplers shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Directional couplers, preserved as specified in 5.1, shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. The requirements for box closure, waterproofing, and reinforcing shall be in accordance with method V of the appendix to PPP-B-636.

5.2.2 Level C. Directional couplers, preserved as specified in 5.1, shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

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5.2.3 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal to 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.3.1 Level A. Directional couplers, packed as specified in 5.2.1, shall be unitized on pallets in accordance with MIL-STD-147, load type I, with a wood cap (storage aid 5) positioned over each load.

5.2.3.2 Level B. Directional couplers, packed as specified in 5.2.2, shall be unitized as specified in 5.2.4.1, except that weather-resistant fiberboard caps (storage aid 4) shall be used in lieu of wood caps.

5.2.3.3 Level C. Directional couplers, packed as specified in 5.2.3, shall be unitized as specified in MIL-STD-794, except that conformance to MIL-STD-147 is not required.

5.3 Marking. In addition to any special or other identification marking required by the contract (see 6.2), each unit and exterior container and unitized load shall be marked in accordance with MIL-STD-129. The complete military or contractor's type or part number, as applicable (including the Commercial and Government Entity (CAGE)), shall be marked on all unit and supplementary packs in accordance with the identification marking provisions of MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.6.2.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

6.1.1 Directional couplers. Directional couplers may be used in conjunction with a radio-frequency source for injecting a radio-frequency wave into a transmission line so that it flows in one direction only. Directional couplers may also be used to sample a radio-frequency wave flowing in a particular direction in a transmission line while accepting relatively insignificant portions of a radio-frequency wave flowing in the opposite direction.

6.1.2 Packaging requirements. The preservation, packing, and marking herein are intended for direct shipments to the Government. However, at the option of the contractor, or when so specified, the packaging provisions herein are also applicable for the preparation of directional couplers for shipment from the part contractor to the original equipment manufacturer.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Title, number, and date of applicable specification sheet and complete part number (see 3.1 and 1.2).

d. For naval applications, whether magnesium may be used in fabricating directional couplers (see 3.3.6).

e. Whether weatherproof caps are required (see 3.4.15).

f. Special marking required (see 3.21).

g. Specify when alternate test for effective directivity is required (see 4.7.5.1).

h. Continuous wave power rating of primary line (see 4.7.8).

i. Levels of preservation and packing required (see 5.1 and 5.2).

j. If special or additional identification marking is required (see 5.3).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time of award of contract, qualified for inclusion in applicable qualified products list whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the qualified products list is the Space and Naval Warfare Systems Command, SPAWAR 003-121, Department of the Navy, Washington, DC 20363-5100; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC-E), Dayton, OH 45444-5288.

6.3.1 Copies of "Provisions Governing Qualification" SD-6 may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 Definitions. For the purpose of this specification, the following definitions should apply.

6.4.1 Directional coupler. A directional coupler is a transmission-line component characterized physically by two (or three, for certain bidirectional couplers) juxtaposed transmission lines and an associated coupling structure through which a transfer of RF energy from one to the other is effected; its electrical behavior is characterized ideally by such interaction between the two lines that excitation in a single direction in either line produces a response in the companion line in one direction only.

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6.4.1.1 Unidirectional coupler. A unidirectional coupler is a directional coupler so designed as to provide a nominal response in the secondary line to propagation in the primary line, in one direction only.

6.4.1.2 Bidirectional coupler. A bidirectional coupler is a directional coupler so designed as to provide separate and simultaneous nominal reponses in the secondary lines to each of the two directions of propagation in the primary line.

6.4.2 Insertion loss. The loss produced by adding (inserting) a device into a signal transmission path.

6.4.3 Primary line. The primary line of a directional coupler is the line designed to receive the principal flow of RF energy in the transmission line to which the directional coupler is adjunct. The input end of the primary line of a unidirectional coupler is the end into which power must flow in order to produce the maximum power at the output of the secondary line. The other end is called the output (antenna) end.

6.4.4 Secondary line. The secondary line of a directional coupler is the line that is coupled to the primary line by means of the coupling structure. In a unidirectional coupler there is one secondary line. In a bidirectional coupler the secondary lines have a nominal response to a different direction of propagation in the primary line. In some bidirectional couplers the secondary lines are physically separate; in others they run physically together.

6.4.5 Mid-frequency. The mid-frequency of a directional coupler is defined as the arithmetic mean of the limits of its frequency range.

6.5 Conditions for use of level B preservation. When level B preservation is specified (see 5.1.2), this level of protection should be reserved for the acquisition of directional couplers for resupply worldwide under known favorable handling, transportation, and storage conditions.

6.6 Subject term (key word) listing.

Couplers, bidirectional
Couplers, unidirectional
Line, primary
Line, secondary

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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APPENDIX

GROUP QUALIFICATION

10. SCOPE.

10.1 Scope. This appendix provides manufacturers a grouping that can be used to obtain qualification for a number of items by qualifying four items in a group. The grouping shall be in accordance with table VI. Manufacturers may qualify at a lower frequency range than what is listed under the characteristics of the qualifying couplers column of the table. This will qualify couplers at this frequency range with the stated characteristics for a group down to the lowest specified frequency range in the table. Manufacturers may also qualify at a higher coupling variation than stated in the table, this will qualify couplers with this coupling variation value and higher values.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

TABLE VI. Grouping for qualification.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
1	Waveguide Directivity 40 dB Screened Frequency range 18-26.5 GHz Coupling 3 dB Coupling variation ±0.2 dB	Waveguide Directivity 40 dB, 30 dB, and 20 dB Screened and nonscreened Frequency range 18-26.5 GHz to 2.60-3.95 GHz Coupling 3 dB through 10 dB Coupling variation ±0.2 dB to ±1.25 dB
2	Waveguide Directivity 40 dB Screened Frequency range 18-26.5 GHz Coupling 11 dB Coupling variation ±0.2 dB	Waveguide Directivity 40 dB, 30 dB, and 20 dB Screened and nonscreened Frequency range 18-26.5 GHz to 2.60-3.95 GHz Coupling 11 dB through 30 dB Coupling variation ±0.2 dB to ±1.25 dB
3	Waveguide Directivity 40 dB Screened Frequency range 18-26.5 GHz Coupling 31 dB Coupling variation ±0.2 dB	Waveguide Directivity 40 dB, 30 dB, and 20 dB Screened and nonscreened Frequency range 18-26.5 GHz to 2.60-3.95 GHz Coupling 31 dB through 60 dB Coupling variation ±0.2 dB to ±1.25 dB
4	Waveguide Directivity 30 dB Screened Frequency range 18-26.5 GHz Coupling 3 dB Coupling variation ±0.2 dB	Waveguide Directivity 30 dB and 20 dB Screened and nonscreened Frequency range 18-26.5 GHz to 2.60-3.95 GHz Coupling 3 dB through 10 dB Coupling variation ±0.2 dB to ±1.25 dB
5	Waveguide Directivity 30 dB Screened Frequency range 18-26.5 GHz Coupling 11 dB Coupling variation ±0.2 dB	Waveguide Directivity 30 dB and 20 dB Screened and nonscreened Frequency range 18-26.5 GHz to 2.60-3.95 GHz Coupling 11 dB through 30 dB Coupling variation ±0.2 dB to ±1.25 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
6	Waveguide Directivity 30 dB Screened Frequency range 18-26.5 GHz Coupling 31 dB Coupling variation ±0.2 dB	Waveguide Directivity 30 dB and 20 dB Screened and nonscreened Frequency range 18-26.5 GHz to 2.60-3.95 GHz Coupling 31 dB through 60 dB Coupling variation ±0.2 dB to ±1.25 dB
7	Waveguide. Directivity 20 dB Screened Frequency range 18-26.5 GHz Coupling 3 dB Coupling variation ±0.2 dB	Waveguide Directivity 20 dB Screened and nonscreened Frequency range 18-26.5 GHz to 2.60-3.95 GHz Coupling 3 dB through 10 dB Coupling variation ±0.2 dB to ±1.25 dB
8	Waveguide Directivity 20 dB Screened Frequency range 18-26.5 GHz Coupling 11 dB Coupling variation ±0.2 dB	Waveguide Directivity 20 dB Screened and nonscreened Frequency range 18-26.5 GHz to 2.60-3.95 GHz Coupling 11 dB through 30 dB Coupling variation ±0.2 dB to ±1.25 dB
9	Waveguide Directivity 20 dB Screened Frequency range 18-26.5 GHz Coupling 31 dB Coupling variation ±0.2 dB	Waveguide Directivity 20 dB Screened and nonscreened Frequency range 18-26.5 GHz to 2.60-3.95 GHz Coupling 31 dB through 60 dB Coupling variation ±0.2 dB to ±1.25 dB
10	Waveguide Directivity 40 dB Nonscreened Frequency range 18-26.5 GHz Coupling 3 dB Coupling variation ±0.2 dB	Waveguide Directivity 40 dB, 30 dB, and 20 dB Nonscreened Frequency range 18-26.5 GHz to 2.60-3.95 GHz Coupling 3 dB through 10 dB Coupling variation ±0.2 dB to ±1.25 dB
11	Waveguide Directivity 40 dB Nonscreened Frequency range 18-26.5 GHz Coupling 11 dB Coupling variation ±0.2 dB	Waveguide Directivity 40 dB, 30 dB, and 20 dB Nonscreened Frequency range 18-26.5 GHz to 2.60-3.95 GHz Coupling 11 dB through 30 dB Coupling variation ±0.2 dB to ±1.25 dB
12	Waveguide Directivity 40 dB Nonscreened Frequency range 18-26.5 GHz Coupling 31 dB Coupling variation ±0.2 dB	Waveguide Directivity 40 dB, 30 dB, and 20 dB Nonscreened Frequency range 18-26.5 GHz to 2.60-3.95 GHz Coupling 31 dB through 60 dB Coupling variation ±0.2 dB to ±1.25 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
13	Waveguide Directivity 30 dB Nonscreened Frequency range 18-26.5 GHz Coupling 3 dB Coupling variation ±0.2 dB	Waveguide Directivity 30 dB and 20 dB Nonscreened Frequency range 18-26.5 GHz to 2.60-3.95 GHz Coupling 3 dB through 10 dB Coupling variation ±0.2 dB to ±1.25 dB
14	Waveguide Directivity 30 dB Nonscreened Frequency range 18-26.5 GHz Coupling 11 dB Coupling variation ±0.2 dB	Waveguide Directivity 30 dB and 20 dB Nonscreened Frequency range 18-26.5 GHz to 2.60-3.95 GHz Coupling 11 dB through 30 dB Coupling variation ±0.2 dB to ±1.25 dB
15	Waveguide Directivity 30 dB Nonscreened Frequency range 18-26.5 GHz Coupling 31 dB Coupling variation ±0.2 dB	Waveguide Directivity 30 dB and 20 dB Nonscreened Frequency range 18-26.5 GHz to 2.60-3.95 GHz Coupling 31 dB through 60 dB Coupling variation ±0.2 dB to ±1.25 dB
16	Waveguide Directivity 20 dB Nonscreened Frequency range 18-26.5 GHz Coupling 3 dB Coupling variation ±0.2 dB	Waveguide Directivity 20 dB Nonscreened Frequency range 18-26.5 GHz to 2.60-3.95 GHz Coupling 3 dB through 10 dB Coupling variation ±0.2 dB to ±1.25 dB
17	Waveguide Directivity 20 dB Nonscreened Frequency range 18-26.5 GHz Coupling 11 dB Coupling variation ±0.2 dB	Waveguide Directivity 20 dB Nonscreened Frequency range 18-26.5 GHz to 2.60-3.95 GHz Coupling 11 dB through 30 dB Coupling variation ±0.2 dB to ±1.25 dB
18	Waveguide Directivity 20 dB Nonscreened Frequency range 18-26.5 GHz Coupling 31 dB Coupling variation ±0.2 dB	Waveguide Directivity 20 dB Nonscreened Frequency range 18-26.5 GHz to 2.60-3.95 GHz Coupling 31 dB through 60 dB Coupling variation ±0.2 dB to ±1.25 dB
19	Waveguide Directivity 30 dB Screened Frequency range 220-325 GHz Coupling 3 dB Coupling variation ±0.75 dB	Waveguide Directivity 30 dB, 20 dB, and 10 dB Screened and nonscreened Frequency range 220-325 GHz to 22-33 GHz Coupling 3 dB through 10 dB Coupling variation ±0.75 dB to ±1.5 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
20	Waveguide Directivity 30 dB Screened Frequency range 220-325 GHz Coupling 11 dB Coupling variation ±0.75 dB	Waveguide Directivity 30 dB, 20 dB, and 10 dB Screened and nonscreened Frequency range 220-325 GHz to 22-33 GHz Coupling 11 dB through 30 dB Coupling variation ±0.75 dB to ±1.5 dB
21	Waveguide Directivity 30 dB Screened Frequency range 220-325 GHz Coupling 31 dB Coupling variation ±0.75 dB	Waveguide Directivity 30 dB, 20 dB, and 10 dB Screened and nonscreened Frequency range 220-325 GHz to 22-33 GHz Coupling 31 dB through 60 dB Coupling variation ±0.75 dB to ±1.5 dB
22	Waveguide Directivity 20 dB Screened Frequency range 220-325 GHz Coupling 3 dB Coupling variation ±0.75 dB	Waveguide Directivity 20 dB and 10 dB Screened and nonscreened Frequency range 220-325 GHz to 22-33 GHz Coupling 3 dB through 10 dB Coupling variation ±0.75 dB to ±1.5 dB
23	Waveguide Directivity 20 dB Screened Frequency range 220-325 GHz Coupling 11 dB Coupling variation ±0.75 dB	Waveguide Directivity 20 dB and 10 dB Screened and nonscreened Frequency range 220-325 GHz to 22-33 GHz Coupling 11 dB through 30 dB Coupling variation ±0.75 dB to ±1.5 dB
24	Waveguide Directivity 20 dB Screened Frequency range 220-325 GHz Coupling 31 dB Coupling variation ±0.75 dB	Waveguide Directivity 20 dB and 10 dB Screened and nonscreened Frequency range 220-325 GHz to 22-33 GHz Coupling 31 dB through 60 dB Coupling variation ±0.75 dB to ±1.5 dB
25	Waveguide Directivity 10 dB Screened Frequency range 220-325 GHz Coupling 3 dB Coupling variation ±0.75 dB	Waveguide Directivity 10 dB Screened and nonscreened Frequency range 220-325 GHz to 22-33 GHz Coupling 3 dB through 10 dB Coupling variation ±0.75 dB to ±1.5 dB
26	Waveguide Directivity 10 dB Screened Frequency range 220-325 GHz Coupling 11 dB Coupling variation ±0.75 dB	Waveguide Directivity 10 dB Screened and nonscreened Frequency range 220-325 GHz to 22-33 GHz Coupling 11 dB through 30 dB Coupling variation ±0.75 dB to ±1.5 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
27	Waveguide Directivity 10 dB Screened Frequency range 220-325 GHz Coupling 31 dB Coupling variation ±0.75 dB	Waveguide Directivity 10 dB Screened and nonscreened Frequency range 220-325 GHz to 22-33 GHz Coupling 31 dB through 60 dB Coupling variation ±0.75 dB to ±1.5 dB
28	Waveguide Directivity 30 dB Nonscreened Frequency range 220-325 GHz Coupling 3 dB Coupling variation ±0.75 dB	Waveguide Directivity 30 dB, 20 dB, and 10 dB Nonscreened Frequency range 220-325 GHz to 22-33 GHz Coupling 3 dB through 10 dB Coupling variation ±0.75 dB to ±1.5 dB
29	Waveguide Directivity 30 dB Nonscreened Frequency range 220-325 GHz Coupling 11 dB Coupling variation ±0.75 dB	Waveguide Directivity 30 dB, 20 dB, and 10 dB Nonscreened Frequency range 220-325 GHz to 22-33 GHz Coupling 11 dB through 30 dB Coupling variation ±0.75 dB to ±1.5 dB
30	Waveguide Directivity 30 dB Nonscreened Frequency range 220-325 GHz Coupling 31 dB Coupling variation ±0.75 dB	Waveguide Directivity 30 dB, 20 dB, and 10 dB Nonscreened Frequency range 220-325 GHz to 22-33 GHz Coupling 31 dB through 60 dB Coupling variation ±0.75 dB to ±1.5 dB
31	Waveguide Directivity 20 dB Nonscreened Frequency range 220-325 GHz Coupling 3 dB Coupling variation ±0.75 dB	Waveguide Directivity 20 dB and 10 dB Nonscreened Frequency range 220-325 GHz to 22-33 GHz Coupling 3 dB through 10 dB Coupling variation ±0.75 dB to ±1.5 dB
32	Waveguide Directivity 20 dB Nonscreened Frequency range 220-325 GHz Coupling 11 dB Coupling variation ±0.75 dB	Waveguide Directivity 20 dB and 10 dB Nonscreened Frequency range 220-325 GHz to 22-33 GHz Coupling 11 dB through 30 dB Coupling variation ±0.75 dB to ±1.5 dB
33	Waveguide Directivity 20 dB Nonscreened Frequency range 220-325 GHz Coupling 31 dB Coupling variation ±0.75 dB	Waveguide Directivity 20 dB and 10 dB Nonscreened Frequency range 220-325 GHz to 22-33 GHz Coupling 31 dB through 60 dB Coupling variation ±0.75 dB to ±1.5 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
34	Waveguide Directivity 10 dB Nonscreened Frequency range 220-325 GHz Coupling 3 dB Coupling variation ±0.75 dB	Waveguide Directivity 10 dB Nonscreened Frequency range 220-325 GHz to 22-33 GHz Coupling 3 dB through 10 dB Coupling variation ±0.75 dB to ±1.5 dB
35	Waveguide Directivity 10 dB Nonscreened Frequency range 220-325 GHz Coupling 11 dB Coupling variation ±0.75 dB	Waveguide Directivity 10 dB Nonscreened Frequency range 220-325 GHz to 22-33 GHz Coupling 11 dB through 30 dB Coupling variation ±0.75 dB to ±1.5 dB
36	Waveguide Directivity 10 dB Nonscreened Frequency range 220-325 GHz Coupling 31 dB Coupling variation ±0.75 dB	Waveguide Directivity 10 dB Nonscreened Frequency range 220-325 GHz to 22-33 GHz Coupling 31 dB through 60 dB Coupling variation ±0.75 dB to ±1.5 dB
37	Coaxial Directivity 30 dB Screened Frequency range 100 kHz-46 GHz Coupling 3 dB Coupling variation ±0.7 dB	Coaxial Directivity 30 dB, 25 dB, 20 dB, and 10 dB Screened and nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 3 dB through 10 dB Coupling variation ±0.7 dB to ±1.2 dB
38	Coaxial Directivity 30 dB Screened Frequency range 100 kHz-46 GHz Coupling 11 dB Coupling variation ±0.7 dB	Coaxial Directivity 30 dB, 25 dB, 20 dB, and 10 dB Screened and nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 11 dB through 30 dB Coupling variation ±0.7 dB to ±1.2 dB
39	Coaxial Directivity 30 dB Screened Frequency range 100 kHz-46 GHz Coupling 31 dB Coupling variation ±0.7 dB	Coaxial Directivity 30 dB, 25 dB, 20 dB, and 10 dB Screened and nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 31 dB through 40 dB Coupling variation ±0.7 dB to ±1.2 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
40	Coaxial Directivity 25 dB Screened Frequency range 100 kHz-46 GHz Coupling 3 dB Coupling variation ±0.7 dB	Coaxial Directivity 25 dB, 20 dB, and 10 dB Screened and nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 3 dB through 10 dB Coupling variation ±0.7 dB to ±1.2 dB
41	Coaxial Directivity 25 dB Screened Frequency range 100 kHz-46 GHz Coupling 11 dB Coupling variation ±0.7 dB	Coaxial Directivity 25 dB, 20 dB, and 10 dB Screened and nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 11 dB through 30 dB Coupling variation ±0.7 dB to ±1.2 dB
42	Coaxial Directivity 25 dB Screened Frequency range 100 kHz-46 GHz Coupling 31 dB Coupling variation ±0.7 dB	Coaxial Directivity 25 dB, 20 dB, and 10 dB Screened and nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 31 dB through 40 dB Coupling variation ±0.7 dB to ±1.2 dB
43	Coaxial Directivity 20 dB Screened Frequency range 100 kHz-46 GHz Coupling 3 dB Coupling variation ±0.7 dB	Coaxial Directivity 20 dB and 10 dB Screened and nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 3 dB through 10 dB Coupling variation ±0.7 dB to ±1.2 dB
44	Coaxial Directivity 20 dB Screened Frequency range 100 kHz-46 GHz Coupling 11 dB Coupling variation ±0.7 dB	Coaxial Directivity 20 dB and 10 dB Screened and nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 11 dB through 30 dB Coupling variation ±0.7 dB to ±1.2 dB
45	Coaxial Directivity 20 dB Screened Frequency range 100 kHz-46 GHz Coupling 31 dB Coupling variation ±0.7 dB	Coaxial Directivity 20 dB and 10 dB Screened and nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 31 dB through 40 dB Coupling variation ±0.7 dB to ±1.2 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
46	Coaxial Directivity 10 dB Screened Frequency range 100 kHz-46 GHz Coupling 3 dB Coupling variation ±0.7 dB	Coaxial Directivity 10 dB Screened and nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 3 dB through 10 dB Coupling variation ±0.7 dB to ±1.2 dB
47	Coaxial Directivity 10 dB Screened Frequency range 100 kHz-46 GHz Coupling 11 dB Coupling variation ±0.7 dB	Coaxial Directivity 10 dB Screened and nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 11 dB through 30 dB Coupling variation ±0.7 dB to ±1.2 dB
48	Coaxial Directivity 10 dB Screened Frequency range 100 kHz-46 GHz Coupling 31 dB Coupling variation ±0.7 dB	Coaxial Directivity 10 dB Screened and nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 31 dB through 40 dB Coupling variation ±0.7 dB to ±1.2 dB
49	Coaxial Directivity 30 dB Nonscreened Frequency range 100 kHz-46 GHz Coupling 3 dB Coupling variation ±0.7 dB	Coaxial Directivity 30 dB, 25 dB, 20 dB, and 10 dB Nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 3 dB through 10 dB Coupling variation ±0.7 dB to ±1.2 dB
50	Coaxial Directivity 30 dB Nonscreened Frequency range 100 kHz-46 GHz Coupling 11 dB Coupling variation ±0.7 dB	Coaxial Directivity 30 dB, 25 dB, 20 dB, and 10 dB Nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 11 dB through 30 dB Coupling variation ±0.7 dB to ±1.2 dB
51	Coaxial Directivity 30 dB Nonscreened Frequency range 100 kHz-46 GHz Coupling 31 dB Coupling variation ±0.7 dB	Coaxial Directivity 30 dB, 25 dB, 20 dB, and 10 dB Nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 31 dB through 40 dB Coupling variation ±0.7 dB to ±1.2 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
52	Coaxial Directivity 25 dB Nonscreened Frequency range 100 kHz-46 GHz Coupling 3 dB Coupling variation ±0.7 dB	Coaxial Directivity 25 dB, 20 dB, and 10 dB Nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 3 dB through 10 dB Coupling variation ±0.7 dB to ±1.2 dB
53	Coaxial Directivity 25 dB Nonscreened Frequency range 100 kHz-46 GHz Coupling 11 dB Coupling variation ±0.7 dB	Coaxial Directivity 25 dB, 20 dB, and 10 dB Nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 11 dB through 30 dB Coupling variation ±0.7 dB to ±1.2 dB
54	Coaxial Directivity 25 dB Nonscreened Frequency range 100 kHz-46 GHz Coupling 31 dB Coupling variation ±0.7 dB	Coaxial Directivity 25 dB, 20 dB, and 10 dB Nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 31 dB through 40 dB Coupling variation ±0.7 dB to ±1.2 dB
55	Coaxial Directivity 20 dB Nonscreened Frequency range 100 kHz-46 GHz Coupling 3 dB Coupling variation ±0.7 dB	Coaxial Directivity 20 dB and 10 dB Nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 3 dB through 10 dB Coupling variation ±0.7 dB to ±1.2 dB
56	Coaxial Directivity 20 dB Nonscreened Frequency range 100 kHz-46 GHz Coupling 11 dB Coupling variation ±0.7 dB	Coaxial Directivity 20 dB and 10 dB Nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 11 dB through 30 dB Coupling variation ±0.7 dB to ±1.2 dB
57	Coaxial Directivity 20 dB Nonscreened Frequency range 100 kHz-46 GHz Coupling 31 dB Coupling variation ±0.7 dB	Coaxial Directivity 20 dB and 10 dB Nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 31 dB through 40 dB Coupling variation ±0.7 dB to ±1.2 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
58	Coaxial Directivity 10 dB Nonscreened Frequency range 100 kHz-46 GHz Coupling 3 dB Coupling variation ±0.7 dB	Coaxial Directivity 10 dB Nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 3 dB through 10 dB Coupling variation ±0.7 dB to ±1.2 dB
59	Coaxial Directivity 10 dB Nonscreened Frequency range 100 kHz-46 GHz Coupling 11 dB Coupling variation ±0.7 dB	Coaxial Directivity 10 dB Nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 11 dB through 30 dB Coupling variation ±0.7 dB to ±1.2 dB
60	Coaxial Directivity 10 dB Nonscreened Frequency range 100 kHz-46 GHz Coupling 31 dB Coupling variation ±0.7 dB	Coaxial Directivity 10 dB Nonscreened Frequency range 100 kHz-50 GHz to 100 kHz-18.5 GHz Coupling 31 dB through 40 dB Coupling variation ±0.7 dB to ±1.2 dB
61	Coaxial Directivity 30 dB Screened Frequency range 100 kHz-18 GHz Coupling 3 dB Coupling variation ±0.5 dB	Coaxial Directivity 30 dB, 25 dB, 20 dB, and 10 dB Screened and nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 3 dB through 10 dB Coupling variation ±0.5 dB to ±1.0 dB
62	Coaxial Directivity 30 dB Screened Frequency range 100 kHz-18 GHz Coupling 11 dB Coupling variation ±0.5 dB	Coaxial Directivity 30 dB, 25 dB, 20 dB, and 10 dB Screened and nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 11 dB through 30 dB Coupling variation ±0.5 dB to ±1.0 dB
63	Coaxial Directivity 30 dB Screened Frequency range 100 kHz-18 GHz Coupling 31 dB Coupling variation ±0.5 dB	Coaxial Directivity 30 dB, 25 dB, 20 dB, and 10 dB Screened and nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 31 dB through 40 dB Coupling variation ±0.5 dB to ±1.0 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
64	Coaxial Directivity 25 dB Screened Frequency range 100 kHz-18 GHz Coupling 3 dB Coupling variation ±0.5 dB	Coaxial Directivity 25 dB, 20 dB, and 10 dB Screened and nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 3 dB through 10 dB Coupling variation ±0.5 dB to ±1.0 dB
65	Coaxial Directivity 25 dB Screened Frequency range 100 kHz-18 GHz Coupling 11 dB Coupling variation ±0.5 dB	Coaxial Directivity 25 dB, 20 dB, and 10 dB Screened and nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 11 dB through 30 dB Coupling variation ±0.5 dB to ±1.0 dB
66	Coaxial Directivity 25 dB Screened Frequency range 100 kHz-18 GHz Coupling 31 dB Coupling variation ±0.5 dB	Coaxial Directivity 25 dB, 20 dB, and 10 dB Screened and nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 31 dB through 40 dB Coupling variation ±0.5 dB to ±1.0 dB
67	Coaxial Directivity 20 dB Screened Frequency range 100 kHz-18 GHz Coupling 3 dB Coupling variation ±0.5 dB	Coaxial Directivity 20 dB and 10 dB Screened and nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 3 dB through 10 dB Coupling variation ±0.5 dB to ±1.0 dB
68	Coaxial Directivity 20 dB Screened Frequency range 100 kHz-18 GHz Coupling 11 dB Coupling variation ±0.5 dB	Coaxial Directivity 20 dB and 10 dB Screened and nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 11 dB through 30 dB Coupling variation ±0.5 dB to ±1.0 dB
69	Coaxial Directivity 20 dB Screened Frequency range 100 kHz-18 GHz Coupling 31 dB Coupling variation ±0.5 dB	Coaxial Directivity 20 dB and 10 dB Screened and nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 31 dB through 40 dB Coupling variation ±0.5 dB to ±1.0 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
70	Coaxial Directivity 10 dB Screened Frequency range 100 kHz-18 GHz Coupling 3 dB Coupling variation ±0.5 dB	Coaxial Directivity 10 dB Screened and nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 3 dB through 10 dB Coupling variation ±0.5 dB to ±1.0 dB
71	Coaxial Directivity 10 dB Screened Frequency range 100 kHz-18 GHz Coupling 11 dB Coupling variation ±0.5 dB	Coaxial Directivity 10 dB Screened and nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 11 dB through 30 dB Coupling variation ±0.5 dB to ±1.0 dB
72	Coaxial Directivity 10 dB Screened Frequency range 100 kHz-18 GHz Coupling 31 dB Coupling variation ±0.5 dB	Coaxial Directivity 10 dB Screened and nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 31 dB through 40 dB Coupling variation ±0.5 dB to ±1.0 dB
73	Coaxial Directivity 30 dB Nonscreened Frequency range 100 kHz-18 GHz Coupling 3 dB Coupling variation ±0.5 dB	Coaxial Directivity 30 dB, 25 dB, 20 dB, and 10 dB Nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 3 dB through 10 dB Coupling variation ±0.5 dB to ±1.0 dB
74	Coaxial Directivity 30 dB Nonscreened Frequency range 100 kHz-18 GHz Coupling 11 dB Coupling variation ±0.5 dB	Coaxial Directivity 30 dB, 25 dB, 20 dB, and 10 dB Nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 11 dB through 30 dB Coupling variation ±0.5 dB to ±1.0 dB
75	Coaxial Directivity 30 dB Nonscreened Frequency range 100 kHz-18 GHz Coupling 31 dB Coupling variation ±0.5 dB	Coaxial Directivity 30 dB, 25 dB, 20 dB, and 10 dB Nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 31 dB through 40 dB Coupling variation ±0.5 dB to ±1.0 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
76	Coaxial Directivity 25 dB Nonscreened Frequency range 100 kHz-18 GHz Coupling 3 dB Coupling variation ±0.5 dB	Coaxial Directivity 25 dB, 20 dB, and 10 dB Nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 3 dB through 10 dB Coupling variation ±0.5 dB to ±1.0 dB
77	Coaxial Directivity 25 dB Nonscreened Frequency range 100 kHz-18 GHz Coupling 11 dB Coupling variation ±0.5 dB	Coaxial Directivity 25 dB, 20 dB, and 10 dB Nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 11 dB through 30 dB Coupling variation ±0.5 dB to ±1.0 dB
78	Coaxial Directivity 25 dB Nonscreened Frequency range 100 kHz-18 GHz Coupling 31 dB Coupling variation ±0.5 dB	Coaxial Directivity 25 dB, 20 dB, and 10 dB Nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 31 dB through 40 dB Coupling variation ±0.5 dB to ±1.0 dB
79	Coaxial Directivity 20 dB Nonscreened Frequency range 100 kHz-18 GHz Coupling 3 dB Coupling variation ±0.5 dB	Coaxial Directivity 20 dB and 10 dB Nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 3 dB through 10 dB Coupling variation ±0.5 dB to ±1.0 dB
80	Coaxial Directivity 20 dB Nonscreened Frequency range 100 kHz-18 GHz Coupling 11 dB Coupling variation ±0.5 dB	Coaxial Directivity 20 dB and 10 dB Nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 11 dB through 30 dB Coupling variation ±0.5 dB to ±1.0 dB
81	Coaxial Directivity 20 dB Nonscreened Frequency range 100 kHz-18 GHz Coupling 31 dB Coupling variation ±0.5 dB	Coaxial Directivity 20 dB and 10 dB Nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 31 dB through 40 dB Coupling variation ±0.5 dB to ±1.0 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
82	Coaxial Directivity 10 dB Nonscreened Frequency range 100 kHz-18 GHz Coupling 3 dB Coupling variation ±0.5 dB	Coaxial Directivity 10 dB Nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 3 dB through 10 dB Coupling variation ±0.5 dB to ±1.0 dB
83	Coaxial Directivity 10 dB Nonscreened Frequency range 100 kHz-18 GHz Coupling 11 dB Coupling variation ±0.5 dB	Coaxial Directivity 10 dB Nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 11 dB through 30 dB Coupling variation ±0.5 dB to ±1.0 dB
84	Coaxial Directivity 10 dB Nonscreened Frequency range 100 kHz-18 GHz Coupling 31 dB Coupling variation ±0.5 dB	Coaxial Directivity 10 dB Nonscreened Frequency range 100 kHz-18 GHz to 100 kHz-5 MHz Coupling 31 dB through 40 dB Coupling variation ±0.5 dB to ±1.0 dB
85	Flat pack Directivity 30 dB Screened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	Flat pack Directivity 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
86	Flat pack Directivity 30 dB Screened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	Flat pack Directivity 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
87	Flat pack Directivity 25 dB Screened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	Flat pack Directivity 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
88	Flat pack Directivity 25 dB Screened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	Flat pack Directivity 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
89	Flat pack Directivity 20 dB Screened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	Flat pack Directivity 20 dB and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
90	Flat pack Directivity 20 dB Screened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	Flat pack Directivity 20 dB and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
91	Flat pack Directivity 15 dB Screened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	Flat pack Directivity 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
92	Flat pack Directivity 15 dB Screened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	Flat pack Directivity 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
93	Flat pack Directivity 30 dB Screened Frequency range 0.5-1 GHz Coupling 6 dB Coupling variation ±0.5 dB	Flat pack Directivity 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
94	Flat pack Directivity 30 dB Screened Frequency range 0.5-1 GHz Coupling 11 dB Coupling variation ±0.5 dB	Flat pack Directivity 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
95	Flat pack Directivity 25 dB Screened Frequency range 0.5-1 GHz Coupling 6 dB Coupling variation ±0.5 dB	Flat pack Directivity 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
96	Flat pack Directivity 25 dB Screened Frequency range 0.5-1 GHz Coupling 11 dB Coupling variation ±0.5 dB	Flat pack Directivity 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
97	Flat pack Directivity 20 dB Screened Frequency range 0.5-1 GHz Coupling 6 dB Coupling variation ±0.5 dB	Flat pack Directivity 20 dB and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
98	Flat pack Directivity 20 dB Screened Frequency range 0.5-1 GHz Coupling 11 dB Coupling variation ±0.5 dB	Flat pack Directivity 20 dB and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
99	Flat pack Directivity 15 dB Screened Frequency range 0.5-1 GHz Coupling 6 dB Coupling variation ±0.5 dB	Flat pack Directivity 15 dB Screened and nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
100	Flat pack Directivity 15 dB Screened Frequency range 0.5-1 GHz Coupling 11 dB Coupling variation ±0.5 dB	Flat pack Directivity 15 dB Screened and nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
101	Flat pack Directivity 30 dB Nonscreened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	Flat pack Directivity 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
102	Flat pack Directivity 30 dB Nonscreened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	Flat pack Directivity 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
103	Flat pack Directivity 25 dB Nonscreened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	Flat pack Directivity 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
104	Flat pack Directivity 25 dB Nonscreened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	Flat pack Directivity 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
105	Flat pack Directivity 20 dB Nonscreened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	Flat pack Directivity 20 dB and 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
106	Flat pack Directivity 20 dB Nonscreened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	Flat pack Directivity 20 dB and 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
107	Flat pack Directivity 15 dB Nonscreened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	Flat pack Directivity 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
108	Flat pack Directivity 15 dB Nonscreened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	Flat pack Directivity 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
109	Flat pack Directivity 30 dB Nonscreened Frequency range 0.5-1 GHz Coupling 6 dB Coupling variation ±0.5 dB	Flat pack Directivity 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
110	Flat pack Directivity 30 dB Nonscreened Frequency range 0.5-1 GHz Coupling 11 dB Coupling variation ±0.5 dB	Flat pack Directivity 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
111	Flat pack Directivity 25 dB Nonscreened Frequency range 0.5-1 GHz Coupling 6 dB Coupling variation ±0.5 dB	Flat pack Directivity 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
112	Flat pack Directivity 25 dB Nonscreened Frequency range 0.5-1 GHz Coupling 11 dB Coupling variation ±0.5 dB	Flat pack Directivity 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
113	Flat pack Directivity 20 dB Nonscreened Frequency range 0.5-1 GHz Coupling 6 dB Coupling variation ±0.5 dB	Flat pack Directivity 20 dB and 15 dB Nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
114	Flat pack Directivity 20 dB Nonscreened Frequency range 0.5-1 GHz Coupling 11 dB Coupling variation ±0.5 dB	Flat pack Directivity 20 dB and 15 dB Nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
115	Flat pack Directivity 15 dB Nonscreened Frequency range 0.5-1 GHz Coupling 6 dB Coupling variation ±0.5 dB	Flat pack Directivity 15 dB Nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
116	Flat pack Directivity 15 dB Nonscreened Frequency range 0.5-1 GHz Coupling 11 dB Coupling variation ±0.5 dB	Flat pack Directivity 15 dB Nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
117	TO Directivity 30 dB Screened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	TO Directivity 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
118	TO Directivity 30 dB Screened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	TO Directivity 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
119	TO Directivity 30 dB Screened Frequency range 1-2 GHz Coupling 21 dB Coupling variation ±0.5 dB	TO Directivity 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
120	TO Directivity 25 dB Screened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	TO Directivity 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
121	TO Directivity 25 dB Screened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	TO Directivity 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
122	TO Directivity 25 dB Screened Frequency range 1-2 GHz Coupling 21 dB Coupling variation ±0.5 dB	TO Directivity 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
123	TO Directivity 20 dB Screened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	TO Directivity 20 dB and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
124	TO Directivity 20 dB Screened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	TO Directivity 20 dB and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
125	TO Directivity 20 dB Screened Frequency range 1-2 GHz Coupling 21 dB Coupling variation ±0.5 dB	TO Directivity 20 dB and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
126	TO Directivity 15 dB Screened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	TO Directivity 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
127	TO Directivity 15 dB Screened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	TO Directivity 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
128	TO Directivity 15 dB Screened Frequency range 1-2 GHz Coupling 21 dB Coupling variation ±0.5 dB	TO Directivity 15 dB Screened and nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
129	TO Directivity 30 dB Screened Frequency range 0.5-1 GHz Coupling 6 dB Coupling variation ±0.5 dB	TO Directivity 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
130	TO Directivity 30 dB Screened Frequency range 0.5-1 GHz Coupling 11 dB Coupling variation ±0.5 dB	TO Directivity 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
131	TO Directivity 30 dB Screened Frequency range 0.5-1 GHz Coupling 21 dB Coupling variation ±0.5 dB	TO Directivity 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
132	TO Directivity 25 dB Screened Frequency range 0.5-1 GHz Coupling 6 dB Coupling variation ±0.5 dB	TO Directivity 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
133	TO Directivity 25 dB Screened Frequency range 0.5-1 GHz Coupling 11 dB Coupling variation ±0.5 dB	TO Directivity 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
134	TO Directivity 25 dB Screened Frequency range 0.5-1 GHz Coupling 21 dB Coupling variation ±0.5 dB	TO Directivity 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
135	TO Directivity 20 dB Screened Frequency range 0.5-1 GHz Coupling 6 dB Coupling variation ±0.5 dB	TO Directivity 20 dB and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
136	TO Directivity 20 dB Screened Frequency range 0.5-1 GHz Coupling 11 dB Coupling variation ±0.5 dB	TO Directivity 20 dB and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
137	TO Directivity 20 dB Screened Frequency range 0.5-1 GHz Coupling 21 dB Coupling variation ±0.5 dB	TO Directivity 20 dB and 15 dB Screened and nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
138	TO Directivity 15 dB Screened Frequency range 0.5-1 GHz Coupling 6 dB Coupling variation ±0.5 dB	TO Directivity 15 dB Screened and nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
139	TO Directivity 15 dB Screened Frequency range 0.5-1 GHz Coupling 11 dB Coupling variation ±0.5 dB	TO Directivity 15 dB Screened and nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
140	TO Directivity 15 dB Screened Frequency range 0.5-1 GHz Coupling 21 dB Coupling variation ±0.5 dB	TO Directivity 15 dB Screened and nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
141	TO Directivity 30 dB Nonscreened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	TO Directivity 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
142	TO Directivity 30 dB Nonscreened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	TO Directivity 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
143	TO Directivity 30 dB Nonscreened Frequency range 1-2 GHz Coupling 21 dB Coupling variation ±0.5 dB	TO Directivity 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
144	TO Directivity 25 dB Nonscreened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	TO Directivity 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
145	TO Directivity 25 dB Nonscreened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	TO Directivity 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
146	TO Directivity 25 dB Nonscreened Frequency range 1-2 GHz Coupling 21 dB Coupling variation ±0.5 dB	TO Directivity 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
147	TO Directivity 20 dB Nonscreened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	TO Directivity 20 dB and 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
148	TO Directivity 20 dB Nonscreened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	TO Directivity 20 dB and 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
149	TO Directivity 20 dB Nonscreened Frequency range 1-2 GHz Coupling 21 dB Coupling variation ±0.5 dB	TO Directivity 20 dB and 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
150	TO Directivity 15 dB Nonscreened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	TO Directivity 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
151	TO Directivity 15 dB Nonscreened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	TO Directivity 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
152	TO Directivity 15 dB Nonscreened Frequency range 1-2 GHz Coupling 21 dB Coupling variation ±0.5 dB	TO Directivity 15 dB Nonscreened Frequency range 2-500 MHz to 1-2 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
153	TO Directivity 30 dB Nonscreened Frequency range 0.5-1 GHz Coupling 6 dB Coupling variation ±0.5 dB	TO Directivity 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
154	TO Directivity 30 dB Nonscreened Frequency range 0.5-1 GHz Coupling 11 dB Coupling variation ±0.5 dB	TO Directivity 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
155	TO Directivity 30 dB Nonscreened Frequency range 0.5-1 GHz Coupling 21 dB Coupling variation ±0.5 dB	TO Directivity 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
156	TO Directivity 25 dB Nonscreened Frequency range 0.5-1 GHz Coupling 6 dB Coupling variation ±0.5 dB	TO Directivity 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
157	TO Directivity 25 dB Nonscreened Frequency range 0.5-1 GHz Coupling 11 dB Coupling variation ±0.5 dB	TO Directivity 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
158	TO Directivity 25 dB Nonscreened Frequency range 0.5-1 GHz Coupling 21 dB Coupling variation ±0.5 dB	TO Directivity 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
159	TO Directivity 20 dB Nonscreened Frequency range 0.5-1 GHz Coupling 6 dB Coupling variation ±0.5 dB	TO Directivity 20 dB and 15 dB Nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
160	TO Directivity 20 dB Nonscreened Frequency range 0.5-1 GHz Coupling 11 dB Coupling variation ±0.5 dB	TO Directivity 20 dB and 15 dB Nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
161	TO Directivity 20 dB Nonscreened Frequency range 0.5-1 GHz Coupling 21 dB Coupling variation ±0.5 dB	TO Directivity 20 dB and 15 dB Nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
162	TO Directivity 15 dB Nonscreened Frequency range 0.5-1 GHz Coupling 6 dB Coupling variation ±0.5 dB	TO Directivity 15 dB Nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
163	TO Directivity 15 dB Nonscreened Frequency range 0.5-1 GHz Coupling 11 dB Coupling variation ±0.5 dB	TO Directivity 15 dB Nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
164	TO Directivity 15 dB Nonscreened Frequency range 0.5-1 GHz Coupling 21 dB Coupling variation ±0.5 dB	TO Directivity 15 dB Nonscreened Frequency range 2-500 MHz to 0.5-1 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
165	Plug-in Directivity 35 dB Screened Frequency range 2-4 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 35 dB, 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
166	Plug-in Directivity 35 dB Screened Frequency range 2-4 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 35 dB, 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
167	Plug-in Directivity 35 dB Screened Frequency range 2-4 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 35 dB, 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
168	Plug-in Directivity 30 dB Screened Frequency range 2-4 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
169	Plug-in Directivity 30 dB Screened Frequency range 2-4 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
170	Plug-in Directivity 30 dB Screened Frequency range 2-4 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
171	Plug-in Directivity 25 dB Screened Frequency range 2-4 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
172	Plug-in Directivity 25 dB Screened Frequency range 2-4 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
173	Plug-in Directivity 25 dB Screened Frequency range 2-4 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
174	Plug-in Directivity 20 dB Screened Frequency range 2-4 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 20 dB and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
175	Plug-in Directivity 20 dB Screened Frequency range 2-4 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 20 dB and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
176	Plug-in Directivity 20 dB Screened Frequency range 2-4 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 20 dB and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
177	Plug-in Directivity 15 dB Screened Frequency range 2-4 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
178	Plug-in Directivity 15 dB Screened Frequency range 2-4 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
179	Plug-in Directivity 15 dB Screened Frequency range 2-4 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
180	Plug-in Directivity 35 dB Screened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 35 dB, 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
181	Plug-in Directivity 35 dB Screened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 35 dB, 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
182	Plug-in Directivity 35 dB Screened Frequency range 1-2 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 35 dB, 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
183	Plug-in Directivity 30 dB Screened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
184	Plug-in Directivity 30 dB Screened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
185	Plug-in Directivity 30 dB Screened Frequency range 1-2 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
186	Plug-in Directivity 25 dB Screened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
187	Plug-in Directivity 25 dB Screened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
188	Plug-in Directivity 25 dB Screened Frequency range 1-2 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
189	Plug-in Directivity 20 dB Screened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 20 dB and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
190	Plug-in Directivity 20 dB Screened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 20 dB and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
191	Plug-in Directivity 20 dB Screened Frequency range 1-2 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 20 dB and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
192	Plug-in Directivity 15 dB Screened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
193	Plug-in Directivity 15 dB Screened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
194	Plug-in Directivity 15 dB Screened Frequency range 1-2 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
195	Plug-in Directivity 35 dB Screened Frequency range 0.5-500 MHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 35 dB, 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
196	Plug-in Directivity 35 dB Screened Frequency range 0.5-500 MHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 35 dB, 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
197	Plug-in Directivity 35 dB Screened Frequency range 0.5-500 MHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 35 dB, 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
198	Plug-in Directivity 30 dB Screened Frequency range 0.5-500 MHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
199	Plug-in Directivity 30 dB Screened Frequency range 0.5-500 MHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
200	Plug-in Directivity 30 dB Screened Frequency range 0.5-500 MHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 30 dB, 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
201	Plug-in Directivity 25 dB Screened Frequency range 0.5-500 MHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
202	Plug-in Directivity 25 dB Screened Frequency range 0.5-500 MHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
203	Plug-in Directivity 25 dB Screened Frequency range 0.5-500 MHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 25 dB, 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
204	Plug-in Directivity 20 dB Screened Frequency range 0.5-500 MHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
205	Plug-in Directivity 20 dB Screened Frequency range 0.5-500 MHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
206	Plug-in Directivity 20 dB Screened Frequency range 0.5-500 MHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 20 dB, and 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
207	Plug-in Directivity 15 dB Screened Frequency range 0.5-500 MHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
208	Plug-in Directivity 15 dB Screened Frequency range 0.5-500 MHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
209	Plug-in Directivity 15 dB Screened Frequency range 0.5-500 MHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 15 dB Screened and nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
210	Plug-in Directivity 35 dB Nonscreened Frequency range 2-4 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 35 dB, 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
211	Plug-in Directivity 35 dB Nonscreened Frequency range 2-4 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 35 dB, 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
212	Plug-in Directivity 35 dB Nonscreened Frequency range 2-4 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 35 dB, 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
213	Plug-in Directivity 30 dB Nonscreened Frequency range 2-4 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
214	Plug-in Directivity 30 dB Nonscreened Frequency range 2-4 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
215	Plug-in Directivity 30 dB Nonscreened Frequency range 2-4 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
216	Plug-in Directivity 25 dB Nonscreened Frequency range 2-4 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
217	Plug-in Directivity 25 dB Nonscreened Frequency range 2-4 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
218	Plug-in Directivity 25 dB Nonscreened Frequency range 2-4 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
219	Plug-in Directivity 20 dB Nonscreened Frequency range 2-4 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 20 dB and 15 dB Nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
220	Plug-in Directivity 20 dB Nonscreened Frequency range 2-4 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 20 dB and 15 dB Nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
221	Plug-in Directivity 20 dB Nonscreened Frequency range 2-4 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 20 dB and 15 dB Nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
222	Plug-in Directivity 15 dB Nonscreened Frequency range 2-4 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 15 dB Nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
223	Plug-in Directivity 15 dB Nonscreened Frequency range 2-4 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 15 dB Nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
224	Plug-in Directivity 15 dB Nonscreened Frequency range 2-4 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 15 dB Nonscreened Frequency range 0.5-500 MHz to 2-4 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
225	Plug-in Directivity 35 dB Nonscreened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 35 dB, 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
226	Plug-in Directivity 35 dB Nonscreened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 35 dB, 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
227	Plug-in Directivity 35 dB Nonscreened Frequency range 1-2 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 35 dB, 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
228	Plug-in Directivity 30 dB Nonscreened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
229	Plug-in Directivity 30 dB Nonscreened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
230	Plug-in Directivity 30 dB Nonscreened Frequency range 1-2 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
231	Plug-in Directivity 25 dB Nonscreened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
232	Plug-in Directivity 25 dB Nonscreened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
233	Plug-in Directivity 25 dB Nonscreened Frequency range 1-2 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
234	Plug-in Directivity 20 dB Nonscreened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 20 dB and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
235	Plug-in Directivity 20 dB Nonscreened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 20 dB and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
236	Plug-in Directivity 20 dB Nonscreened Frequency range 1-2 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 20 dB and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
237	Plug-in Directivity 15 dB Nonscreened Frequency range 1-2 GHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 15 dB Nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
238	Plug-in Directivity 15 dB Nonscreened Frequency range 1-2 GHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 15 dB Nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
239	Plug-in Directivity 15 dB Nonscreened Frequency range 1-2 GHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 15 dB Nonscreened Frequency range 0.5-500 MHz to 1-2 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
240	Plug-in Directivity 35 dB Nonscreened Frequency range 0.5-500 MHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 35 dB, 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
241	Plug-in Directivity 35 dB Nonscreened Frequency range 0.5-500 MHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 35 dB, 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
242	Plug-in Directivity 35 dB Nonscreened Frequency range 0.5-500 MHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 35 dB, 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
243	Plug-in Directivity 30 dB Nonscreened Frequency range 0.5-500 MHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
244	Plug-in Directivity 30 dB Nonscreened Frequency range 0.5-500 MHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
245	Plug-in Directivity 30 dB Nonscreened Frequency range 0.5-500 MHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 30 dB, 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
246	Plug-in Directivity 25 dB Nonscreened Frequency range 0.5-500 MHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
247	Plug-in Directivity 25 dB Nonscreened Frequency range 0.5-500 MHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
248	Plug-in Directivity 25 dB Nonscreened Frequency range 0.5-500 MHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 25 dB, 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
249	Plug-in Directivity 20 dB Nonscreened Frequency range 0.5-500 MHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB
250	Plug-in Directivity 20 dB Nonscreened Frequency range 0.5-500 MHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
251	Plug-in Directivity 20 dB Nonscreened Frequency range 0.5-500 MHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 20 dB, and 15 dB Nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB
252	Plug-in Directivity 15 dB Nonscreened Frequency range 0.5-500 MHz Coupling 6 dB Coupling variation ±0.5 dB	Plug-in Directivity 15 dB Nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 6 dB through 11 dB Coupling variation ±0.5 dB to ±0.7 dB

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TABLE VI. Grouping for qualification - Continued.

Group number	Characteristics of qualifying couplers	Characteristics of couplers qualified
253	Plug-in Directivity 15 dB Nonscreened Frequency range 0.5-500 MHz Coupling 11 dB Coupling variation ±0.5 dB	Plug-in Directivity 15 dB Nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 11 dB through 20 dB Coupling variation ±0.5 dB to ±0.7 dB
254	Plug-in Directivity 15 dB Nonscreened Frequency range 0.5-500 MHz Coupling 21 dB Coupling variation ±0.5 dB	Plug-in Directivity 15 dB Nonscreened Frequency range 0.5-500 MHz to 1 GHz Coupling 21 dB through 30 dB Coupling variation ±0.5 dB to ±0.7 dB

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, OS
Air Force - 11, 99
DLA - ES

User activities:
Navy - CG, MC, SH

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5985-1003)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-15370D	Couplers, Directional

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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